

Position Title

Data Engineer

Job Purpose

Role: As a core component of KBZ Bank Analytics center of excellence (ACE) team, this role, in technical stream, will be responsible for the design, development and delivery of data engineering solutions such as data lakes, data warehouses, data pipelines and orchestration of operational processes. Data Engineers work hand-in-hand with IT department / platform vendors / merchant systems / social media platforms.

Responsibilities

Key Responsibilities

- **Data Pipeline Development:** Collaborate cross-functionally to design, implement, and maintain high-performance data pipelines for diverse sources. Ensure seamless data movement, transformation, and integration to support analytics and insights.
- **Data Models, Data Lakes, Data Warehouses and Data Marts:** Develop and optimize data models aligned with business needs. Design and maintain data lakes, data warehouses, purpose-built data marts for efficient querying, reporting, and strategic decision-making across operational and analytical contexts in both on premise and cloud environments with the objective of enabling business users to be self-dependent in data extraction, scenario / hypothesis testing, opportunity formulation.
- **Performance Optimization:** Monitor, identify, and proactively address bottlenecks in data pipelines and platform components of various analytical containers in cost efficient manners. Tune queries and streamline data retrieval and transformation processes for sustained high performance.
- **Data Quality, Security and Compliance:** Implement rigorous data validation and quality checks throughout the pipeline. Collaborate internally to ensure data privacy, security, and adherence to industry regulations and standards. Govern access control of data analytics platform / reports / dashboards / analytical throughput while conforming to information security guidelines.

- Collaboration and Documentation: Engage closely with Power User group, BI Team, data science teams to comprehend data requirements and support their analytics endeavors by enabling their data and platform needs either through available data streams or exploring external data streams (through system analysis). Document data flows, processes, and guides to foster effective communication and knowledge sharing.
- Monitoring & Alerting: Daily monitoring to track pipeline issue, performance, resource utilization, and data flow health. In-time resolving the critical data pipeline issue and informing related parties

Job Requirements

Technical Requirements

- SQL skills (Experience with relational databases like PostgreSQL, MySQL, Oracle is highly valuable)
- Advanced experience in Python and Spark.
- Hands-on experience in ETL and data integration, orchestration tools.
- Good understanding of data modelling, data/software engineering best practices, automation, DevOps.
- Proficiency in programming languages such as Python, Java or Scala is preferable.

Job Requirements

- 3+ years of practical experience or relevant internship in a data-focused role (preferred).
- Bachelor's degree in Computer Science, IT, Mathematics, or a related field (or equivalent practical experience).
- Modern stack: dbt + orchestration (Airflow/Dagster/Prefect), cloud warehouse (Redshift/Postgres).
- Data modelling: fact/dimension tables, slowly changing dimensions, event schemas - and when to break the rules.
- Experience working with cloud infrastructure services like Amazon Web Services is preferred.